Lab Handout for Relaxed Tetris Part 2
This is the second part of “Relaxed Tetris”.

Part 3: Clearing lines
The tests inside BoardTest.Java are still all failing.
You should complete Board.isRowFull and Board.clearFullRows to complete this part.
Complete the code for Board.clearFullRows so that full rows will be removed and new empty rows will be inserted at the top of the board. Keep in mind that multiple rows may be full after a given tetromino has landed. You should use the method Board.clearRow to clear a row.

HARD: To get the full marks for the tutor marking, you need to complete Board.isRowFull and Board.clearFullRows using elegant stream operations. Each method can be completed with a single stream expression, that could (for example) start with either rangeX() or rangeY() and work by using the methods of streams and optionals.
The model solution only uses a single ‘while’ keyword and has no need for other for-if-switch-try, ?: etcetera. The tutors will check for this during the in person marking.
Just writing old style code, with conventional statements will give you some of the marks but not the full amount.

By re-running the tests you should see
Runs 58/58 Errors 26 Failures 2.
You will see that the test suit BoardTest is now all green.

Part 4: All the Tetrominos
Here we will complete the missing code for the O, Z, J and L You should refer to the code of tetrominos I, S and T for guidance.
First complete the code for O. Note that O has no orientations, so it should stay the same when rotated.
By re-running the tests you should see
Runs 58/58 Errors 20 Failures 2.
In particular, you will see that the test suit OTest is now all green. Also TestTetrominoOk.testO should now be passing.
Then write the code for Z. It is possible to obtain a correct implementation for Z by extending S. Such an implementation will receive slightly more marks
(by passing test ZTests.testSingleMethodAdded).
By re-running the tests you should see Runs 58/58 Errors 14 Failures 1. In particular, you will see that the test suit ZTest is now all green. Also TestTetrominoOk.testZ should now be passing.
Finally, taking inspiration from S and Z, you can make J, and then L by extending J and overriding exactly 1 method. Such an implementation will receive slightly more marks
(by passing test LTests.testSingleMethodAdded()).
After implementing J, by re-running the tests you should see
Runs 58/58 Errors 7 Failures 1.
After implementing L, by re-running the tests you should see
Runs 58/58 Errors 0 Failures 0.
Note that just passing all the provided tests may not be sufficient to get full marks.
There will be some hidden tests in the submission system.

Part 5: explain Board.toString
Add a multiline comment at the start of each of Board.toString and Board.printLine, explaining in the details how the code of that method works.
In your comments, use correct terminology. The comments you need to add for part 5 are supposed to be sufficiently verbose to convince the tutor that you have a deep understanding of those two methods, and you will have to discuss it with them during in person marking.
Note that comments in production software should instead be concise and to the point.

Submission
Your lab solution should be submitted electronically via the online submission system, linked from the course homepage. You need to submit a zip with all the needed source files as for the other labs.
Do not include your testing files, only your source files.
The submission system will guide you to submit the right files.
You can submit as many times as you want before the deadline, with no penalty.
The last mark will be saved.

Make sure to press 'run tests'.

If you do not see a clear screen showing your mark, you have not completed the submission.

Note that this is only 50% of the lab marks, the other 50% is obtained by engaging with the Tutors for the in person evaluation. They will ask you questions about this activity and you have to answer with clear good terminology.
Note: Failure to engage with the marking system and the tutors in person marking may cause zero marks to be awarded.
Deadlines
- Automatic Marking: Before Friday of week 9
- Tutor mark: any lab slot of Week8 or Week 10